Simulations

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```
library(here)
```

```
## here() starts at /Users/Shi/Documents/school/4. PhD/Research/1. Thesis/TIRS/TIRS_RProj
```

```
library(rio)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(ggplot2)
```

Data

Mérida Incidence

```
## SINAVE case data, only dengue
sinave_dengue_cases <- here::here("data", "SINAVE_Data",
                      "dengue_abierto.csv") %>%
  rio::import() %>%
  dplyr::as_tibble() %>%
  # only want data for Mérida + Kanasín (050, 041 MUNICIPO_RES)
  filter(MUNICIPIO_RES == 50 | MUNICIPIO_RES == 41) %>%
  # change symptom onset date to match cases dataset (i.e., weeks since
  #  start of ABD season)
  mutate(Week_onset = ymd(FECHA_SIGN_SINTOMAS),
         Week_onset = epiweek(Week_onset),
```

```
        Year = year(FECHA_SIGN_SINTOMAS)) %>%
  relocate(c(FECHA_SIGN_SINTOMAS, Week_onset), .before = ID_REGISTRO) %>%
  filter(ESTATUS_CASO == 2) %>%  # 1 likely; 2 confirmed; 3 discarded
  mutate(age_grp = as.factor(ifelse(EDAD_ANOS < 18, "Child", "Adult")))

# incidence in merida??
city_cases_by_week <- sinave_dengue_cases %>%
  count(Week_onset, name = "cases_per_week")
write.table(city_cases_by_week, file = "city_cases_by_week.txt", sep = " ",
            row.names = FALSE)

## divided by population...
yuc_pop <- read.csv("RESAGEBURB2020 - 31 Yucatán.csv")

I_m <- city_cases_by_week$cases_per_week #/ yuc_pop$POBTOT[1]


#there are a lot of cases at the beginning of the year here...are they part of the '23 outbreak?
# cases for '23 stop at 47 weeks...what about the rest of the year?
sinave_24 <- here::here("data", "SINAVE_Data",
                        "datos_abiertos_dengue_191224.csv") %>%
  rio::import() %>%
  dplyr::as_tibble() %>%
  # only want data for Mérida + Kanasín (050, 041 MUNICIPO_RES)
  filter(MUNICIPIO_RES == 50 | MUNICIPIO_RES == 41) %>%
  # change symptom onset date to match cases dataset (i.e., weeks since
  #  start of ABD season)
  mutate(Week_onset = dmy(FECHA_SIGN_SINTOMAS),
         Week_onset = epiweek(Week_onset),
         Year = year(FECHA_SIGN_SINTOMAS)) %>%
  relocate(c(FECHA_SIGN_SINTOMAS, Week_onset), .before = ID_REGISTRO) %>%
  filter(ESTATUS_CASO == 2) %>%  # 1 likely; 2 confirmed; 3 discarded
  mutate(age_grp = as.factor(ifelse(EDAD_ANOS < 18, "Child", "Adult"))) %>%
  count(Week_onset, name = "cases_per_week")
```

Simulation

Functions

```
set.seed(12345)
```

Trial Population Structure

```
sim_pop <- function(n_clusters = 50, n_per_cluster, n_per_block,
                    min_age = 1, max_age = 82){

  n <- n_clusters * n_per_cluster
```

```
  trial_pop_df <- data.frame(
    "ID" = 1:n,
    "Cluster" = rep(1:n_clusters, each = n_per_cluster),
    "Block" = rep_len(1:round(n_per_cluster/n_per_block, 0), n_clusters),
    "TIRS" = rep(c(0, 1), each = n_per_cluster, length.out = n),
    "Age" = sample(min_age:max_age, n, replace = TRUE)
    )

  trial_pop_df$cluster_block <- paste(trial_pop_df$Cluster,
                                      trial_pop_df$Block, sep = "")
  return(trial_pop_df)
}
```

Infect Population

```
# vector of generation time pmf given the parameters;
# evaluated at each time point
gen_time <- function(Gmu, Gs2, max_gen_time) {
  shape <- Gmu^2 / Gs2
  scale <- Gs2 / Gmu

  i <- 0:max_gen_time

  genProb <- pgamma(i + 1, shape = shape, scale = scale) -
               pgamma(i,     shape = shape, scale = scale)

  return(genProb)
}
```

Generation Time

```
infect_pop <- function(trial_pop, gen_prob, initially_infected = 1,
                       alpha, beta, gamma, r) {

  N <- nrow(trial_pop)
  trial_pop$infected <- 0
  trial_pop$tau <- NA
  trial_pop$infector <- NA

  lambda_df <- data.frame(
    time = integer(),
    person = integer(),
    block = double(),
    cluster = double(),
    city = double(),
    total_hazard = double())
```

```r
# already?? infected??? within the trial???
## initial infections??????? 5 initial???
initial_ids <- sample(1:N, initially_infected)
trial_pop$infected[initial_ids] <- 1
trial_pop$tau[initial_ids] <- 1
trial_pop$index_case <- 0
trial_pop$index_case[initial_ids] <- 1

lambdas <- vector("list", N * trial_end)
counter <- 1

for (t  in 2:trial_end){

  new_infections <- c()

  for (j in 1:N){
    if (trial_pop$infected[j] == 1) { next }

    lambda_jb <- 0
    lambda_jc <- 0
    lambda_jm <- 0

    # block FOI
    same_block <- which(trial_pop$cluster_block ==
                          trial_pop$cluster_block[j] & (1:N) != j)
    for (i in same_block){
      if (trial_pop$infected[i] == 1) {
        dt <- t - trial_pop$tau[i]
        if (dt <= max_gen_time){
          if (trial_pop$TIRS[i] == 1) {
            lambda_jb <- lambda_jb + r * alpha * gen_prob[dt]
            # treatment reduction * cum. hazard from i infecting j * prob that gen time is dt
          } else {
            lambda_jb <- lambda_jb + alpha * gen_prob[dt]
            # cum. hazard from i infecting j * prob that gen time is dt
          }
        }
      }
    }

    # cluster FOI
    same_cluster <- which(trial_pop$Cluster == trial_pop$Cluster[j]
                          & (1:N) != j)
    diff_block <- setdiff(same_cluster, same_block)
    for (i in diff_block) {
      if (trial_pop$infected[i] == 1) {
        dt <- t - trial_pop$tau[i]
        if (dt <= max_gen_time){
          if (trial_pop$TIRS[i] == 1) {
            lambda_jc <- lambda_jc + r * beta * gen_prob[dt]
          } else {
            lambda_jc <- lambda_jc + beta * gen_prob[dt]
            # cum. hazard from i infecting j * prob that gen time is dt
```

```
      }
    }
  }
}

# merida FOI
for (z in 1:(t-1)) {
  dt <- t - z
  if (dt <= max_gen_time) {
    lambda_jm <- lambda_jm + gamma * (I_m[z] / yuc_pop$POBTOT[1]) * gen_prob[t - z]
    # cum. hazard from a case in merid * cases in merida * prob gen time is dt
  }
}

# total FOI on j
lambda_j <- lambda_jb + lambda_jc + lambda_jm

lambdas[[counter]] <- data.frame(time = t,
                                 person = j,
                                 block = lambda_jb,
                                 cluster = lambda_jc,
                                 city = lambda_jm,
                                 total_hazard = lambda_j
                                )
counter <- counter + 1

# probability of infection given hazard
p_j <- 1 - exp(-lambda_j)

# j becomes a new infection
if (runif(1) < p_j) {
  # possible infectors of j
  sources <- c(same_block, diff_block, 0)
  weights <- c(sapply(same_block, function(i) {
    if (trial_pop$infected[i] == 1) {
      dt <- t - trial_pop$tau[i]
      if (dt <= max_gen_time){
        base <- alpha * gen_prob[dt]
        # hazard from i infecting j * prob that gen time is dt
        if (trial_pop$TIRS[i] == 1) { base <- r * base }
        return(base)
      }
    }
    return(0)
  }), sapply(diff_block, function(i){
    if (trial_pop$infected[i] == 1) {
      dt <- t - trial_pop$tau[i]
      if (dt <= max_gen_time){
        base <- beta * gen_prob[dt]
        if (trial_pop$TIRS[i] == 1) { base <- base * r }
        return(base)
      }
    }
```

```
          return(0)
        }), lambda_jm)

        # pick infector
        if (sum(weights) > 0) {
          infector <- sample(sources, 1, prob = weights)
        } else {
          infector <- 0 # infection is imported
        }

        new_infections <- c(new_infections, j)
        if (infector == 0) {
          trial_pop$infector[j] <- 0 # imported infection
        } else {
          trial_pop$infector[j] <- trial_pop$ID[infector]
        }
      }
    }

    if (length(new_infections) > 0) {
      trial_pop$infected[new_infections] <- 1
      trial_pop$tau[new_infections] <- t
    }
  }
  lambda_df <- do.call(rbind, lambdas[1:(counter-1)])

  return(list(trial_pop = trial_pop, lambda_df = lambda_df))
}
```

Infections

Summary Statistics

of the simulated populations

```
get_summary <- function(infected_pop){
  # epidemic curve compared to real data :')
  incidence_df <- infected_pop$trial_pop %>%
    filter(is.finite(tau)) %>%
    count(tau, name = "cases") %>%
    complete(tau = 1:trial_end, fill = list(cases = 0)) %>%
    rename(time = tau) %>%
    mutate(case_rate = cases / (n_clusters*n_per_cluster))

  real_df <- data.frame(
    time = 1:nrow(city_cases_by_week),
    case_rate = city_cases_by_week$cases_per_week / yuc_pop$POBTOT[1])

  plot <- ggplot() +
    geom_line(data = incidence_df, aes(time, case_rate), color = "blue") +
    geom_line(data = real_df, aes(time, case_rate), color = "red") +
    labs(title = "Simulated (blue) vs Real (red)") +
```

```
    theme_classic()

  # number of cases at the beginning ?
  head <- head(incidence_df, 10)

  # R_t?
  offspring <- table(na.omit(infected_pop$trial_pop$infector))
  Rt_dt <- infected_pop$trial_pop %>%
    filter(!is.na(infector)) %>%
    group_by(time = tau) %>%
    summarize (R_t = round(mean(table(infector)), 2))

  # distribution of secondary cases
  offspring_count <- as.numeric(offspring)

  hist_off <- hist(offspring_count)
  mean_off <- mean(offspring_count)
  var_off <- var(offspring_count)

  # fraction of infections from each source
  infec_frac <- infected_pop$lambda_df %>%
    mutate(total = block + cluster + city) %>%
    summarize(block_frac = round(mean(block / total, na.rm = TRUE), 2),
              cluster_frac = round(mean(cluster / total, na.rm = TRUE), 2),
              city_frac = round(mean(city / total, na.rm = TRUE), 2))

  # attack rate?
  attack_rate <- mean(infected_pop$trial_pop$infected)

  return(list("Epidemic Curve" = plot, "First 10 Infections" = head,
              "Fraction of Infections from Each Source" = infec_frac,
              "Rt over time" = Rt_dt,
              "Histogram of Offspring Distribution" = hist_off,
              "Mean of Offspring Distribution" = mean_off,
              "Variance of Offspring Distribution" = var_off,
              "Attack Rate" = attack_rate))
}
```

1. epidemic curve
2. first few cases
3. R_t, effective reproductive number (average number of secondary cases?)

Parameter Testing

Set Up

```
run_sims <- function(num_sims, trial_pop, gen_prob, initially_infected, alpha,
                     beta, gamma, r, make_plots = TRUE) {

  sim_list <- list()
  offspring_index <- c()
```

```r
rr_peak <- c()
time_diff_at_peaks <- c()
control_list <- list()
tirs_list <- list()
prop_infected <- c()
control_infected <- c()
tirs_infected <- c()

real_df <- data.frame(
  time = 1:nrow(city_cases_by_week),
  case_rate = city_cases_by_week$cases_per_week / yuc_pop$POBTOT[1])

for (s in 1:num_sims) {
  infected_pop <- infect_pop(trial_pop, gen_prob, 1, alpha = alpha,
                             beta = beta, gamma = gamma, r = r)

  # epidemic curve
  ## average incidence across simulations
  incidence <- infected_pop$trial_pop %>%
    filter(is.finite(tau)) %>%
    count(tau, name = "cases") %>%
    complete(tau = 1:trial_end, fill = list(cases = 0)) %>%
    rename(time = tau) %>%
    mutate(case_rate = cases / (n_clusters*n_per_cluster),
           sim = s)

  sim_list[[s]] <- incidence

  ## by trial arm
  control <- infected_pop$trial_pop %>%
    filter(is.finite(tau)) %>%
    filter(TIRS == 0) %>%
    count(tau, name = "cases") %>%
    complete(tau = 1:trial_end, fill = list(cases = 0)) %>%
    rename(time = tau) %>%
    mutate(sim = s)
  control_list[[s]] <- control

  tirs <- infected_pop$trial_pop %>%
    filter(is.finite(tau)) %>%
    filter(TIRS == 1) %>%
    count(tau, name = "cases") %>%
    complete(tau = 1:trial_end, fill = list(cases = 0)) %>%
    rename(time = tau) %>%
    mutate(sim = s)
  tirs_list[[s]] <- tirs

  # count offspring
  index_ids <- infected_pop$trial_pop$ID[infected_pop$trial_pop$index_case == 1]
  count_offspring <- sapply(index_ids, function(id) {
    sum(infected_pop$trial_pop$infector == id, na.rm = TRUE)
  })
  offspring_index[s] <- mean(count_offspring)
```

```r
  # find peak incidence in simulation
  study_peak <- max(incidence$case_rate)
  real_peak <- max(real_df$case_rate)

  rr_peak[s] <- study_peak / real_peak # difference in magnitude
  time_diff_at_peaks[s] <- incidence$time[which.max(incidence$case_rate)] -
    real_df$time[which.max(real_df$case_rate)]

  peak_time <- incidence$time[which.max(incidence$case_rate)]
  real_peak_same_time <- real_df$case_rate[real_df$time == peak_time]
  rr_peak_same_time[s] <- study_peak / real_peak_same_time

  # proportion of population infected at end of season
  prop_infected[s] <- sum(incidence$cases) / (n_clusters*n_per_cluster)

  # num infected by trial arm
  control_infected[s] <- sum(control$cases)
  tirs_infected[s] <- sum(tirs$cases)
}

# epidemic curve
if (make_plots == TRUE) {
  all_sims <- bind_rows(sim_list)
  epi_curve_summ <- all_sims %>%
  group_by(time) %>%
  summarize(mean_rate = mean(case_rate),
            lower = quantile(case_rate, 0.025),
            upper = quantile(case_rate, 0.975))
epi_curve <- ggplot() +
  geom_ribbon(data = epi_curve_summ,
              aes(x = time, ymin = lower, ymax = upper),
              fill = "blue", alpha = 0.2) +
  geom_line(data = epi_curve_summ,
            aes(time, mean_rate),
            color = "blue") +
  geom_line(data = real_df,
            aes(time, case_rate),
            color = "red") +
  labs(title = paste0("Average Case Rate over ", num_sims,
                      " Simulations (Blue) vs. Mérida Incidence Rate")) +
  theme_classic()
all_controls <- bind_rows(control_list)
all_tirs <- bind_rows(tirs_list)
epi_curve_controls <- all_controls %>%
  group_by(time) %>%
  summarize(avg_num_cases = round(mean(cases), 2),
            lower = quantile(avg_num_cases, 0.025),
            upper = quantile(avg_num_cases, 0.975))
epi_curve_tirs <- all_tirs %>%
  group_by(time) %>%
  summarize(avg_num_cases = round(mean(cases), 2),
            lower = quantile(avg_num_cases, 0.025),
            upper = quantile(avg_num_cases, 0.975))
```

```
  epi_curve_arms <- ggplot() +
    geom_ribbon(data = epi_curve_controls,
                aes(x = time, ymin = lower, ymax = upper),
                fill = "lightgreen", alpha = 0.2) +
    geom_line(data = epi_curve_controls,
              aes(time, avg_num_cases),
              color = "lightgreen") +
    geom_ribbon(data = epi_curve_tirs,
                aes(x = time, ymin = lower, ymax = upper),
                fill = "pink", alpha = 0.2) +
    geom_line(data = epi_curve_tirs,
              aes(time, avg_num_cases),
              color = "pink") +
    labs(title = paste0("Average Number of Cases over ", num_sims, " Simulations for Control (Green) \n
    theme_classic()

  hist_offspring <- hist(offspring_index)
  } else {
    epi_curve <- NULL
    epi_curve_arms <- NULL
    hist_offspring <- NULL
  }

  # R_0
  avg_offspring <- round(mean(offspring_index), 2)
  var_offspring <- round(var(offspring_index), 2)

  # rate ratio between study population and merida
  avg_rr_peak <- mean(rr_peak, na.rm = TRUE)
  var_rr_peak <- var(rr_peak, na.rm = TRUE)
  avg_time_diff <- mean(time_diff_at_peaks, na.rm = TRUE)

  # proportion of population infected
  avg_prop_infected <- mean(prop_infected, na.rm = TRUE)

  # avg infected by trial arm
  avg_control_infected <- mean(control_infected, na.rm = TRUE)
  avg_tirs_infected <- mean(tirs_infected, na.rm = TRUE)

  return(list(epi_curve = epi_curve, hist_offspring = hist_offspring,
              "Average R_0" = avg_offspring, "Variance of R_0" = var_offspring,
              "Num Simulations" = num_sims,
              "Average RR at Peak Incidence" = avg_rr_peak,
              "Variance of RR at Peak Incidence" = var_rr_peak,
              "Average Difference in Weeks Between Simulated Study Peak and City Peak"
              = avg_time_diff, epi_curve_arms = epi_curve_arms,
              "Average Proportion of People Infected by End of ABD Season" =
                avg_prop_infected,
              "Average No. Individuals Infected in Control Arm" =
                avg_control_infected,
              "Average No. Individuals Infected in TIRS Arm" =
                avg_tirs_infected))
}
```

Trial Population:

```
min_age <- 1 # youngest age
max_age <- 82 # oldest age

trial_start <- 1 # first week of the year
trial_end <- nrow(city_cases_by_week) # 52 is the last week of the year but we don't have city incidenc

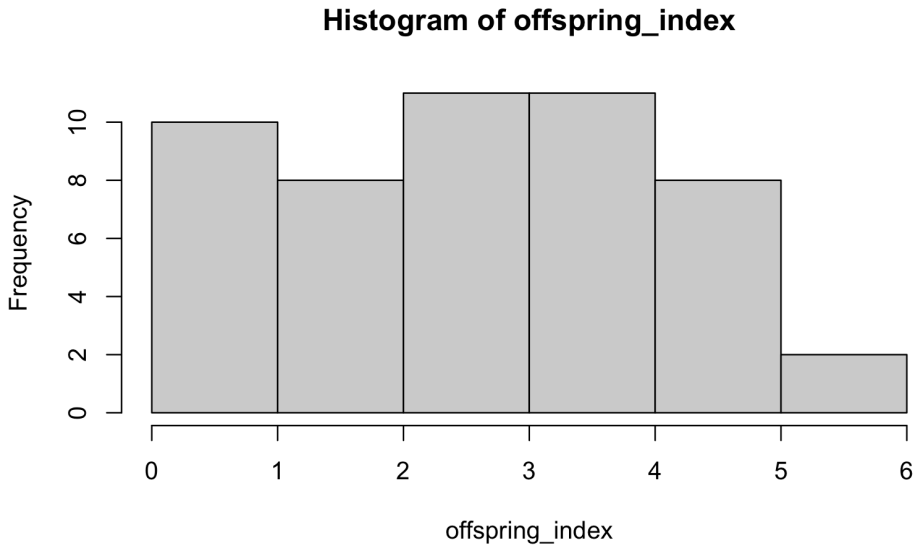
# parameters
Gmu <- 2; Gs2 <- 1; max_gen_time <- 90 # 17 days?; 2 days?
# probability that the generation time = [index]
gen_prob <- gen_time(Gmu, Gs2, max_gen_time)

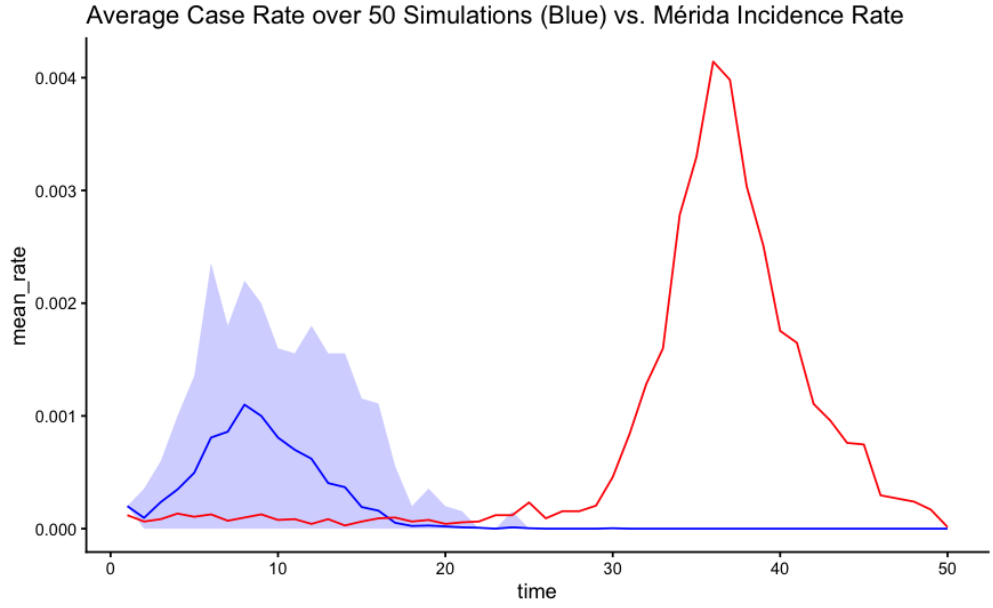
# Population size
n_clusters <- 10; n_per_cluster <- 500; n_per_block <- 50
trial_pop <- sim_pop(n_clusters, n_per_cluster, n_per_block)
```

alpha

```
Sys.time()
alpha <- -log(46/49); beta <- 0; gamma <- 0; r <- 1
run_sims(50, trial_pop, gen_prob, 1, alpha, beta, gamma, r)
Sys.time()
```

```
## $`Average R_0`
## [1] 3
##
## $`Variance of R_0`
## [1] 2.73
##
## $`Num Simulations`
## [1] 50
##
## $`Average RR at Peak Incidence`
## [1] 0.3958843
##
## $`Variance of RR at Peak Incidence`
## [1] 0.02140549
##
## $`Average Difference in Weeks Between Simulated Study Peak and City Peak`
## [1] -27.8
```



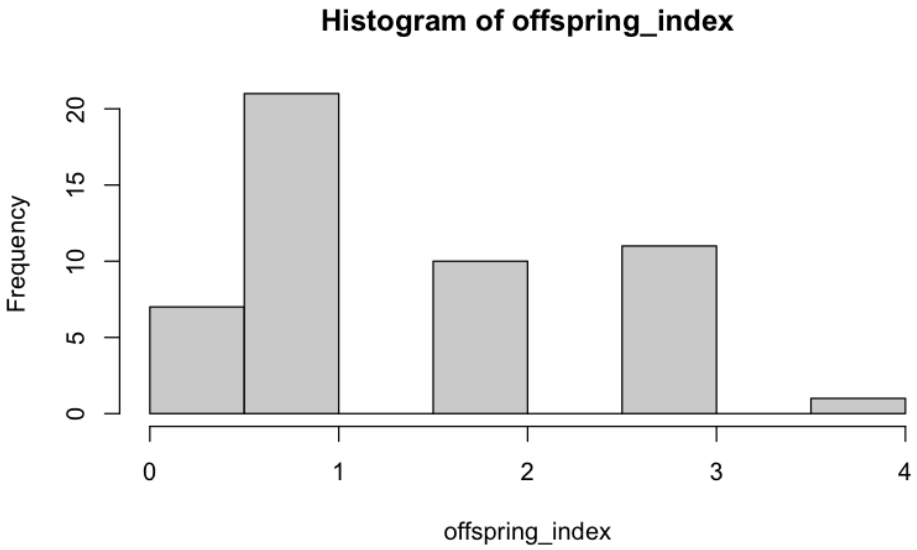


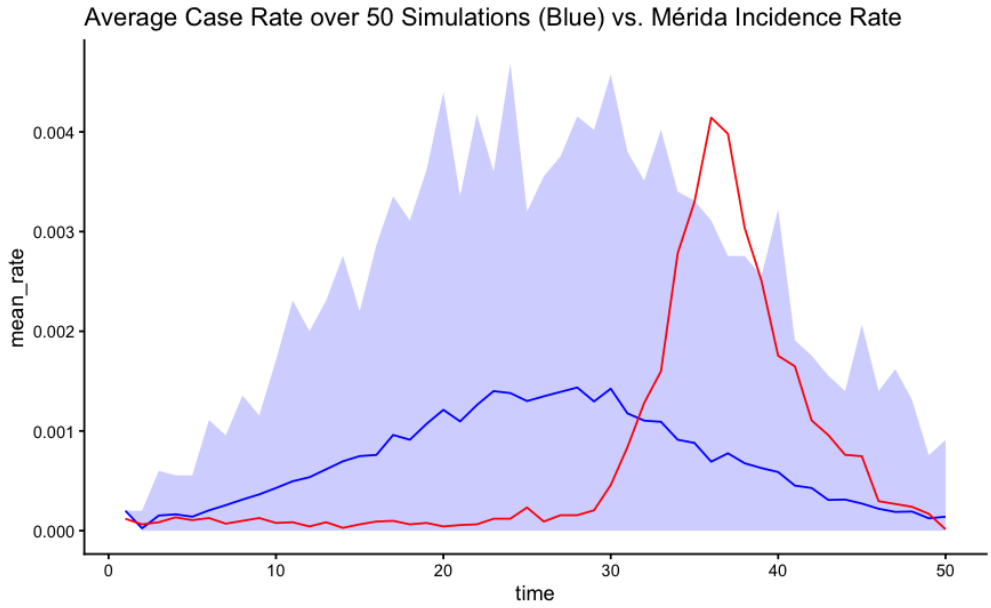
beta

```
Sys.time()
alpha <- 0; beta <- -log(897/900); gamma <- 0; r <- 1
run_sims(50, trial_pop, gen_prob, 1, alpha, beta, gamma, r)
Sys.time()
```

```
## $`Average R_0`
## [1] 1.56
##
## $`Variance of R_0`
```

```
## [1] 1.11
##
## $`Num Simulations`
## [1] 50
##
## $`Average RR at Peak Incidence`
## [1] 0.5938264
##
## $`Variance of RR at Peak Incidence`
## [1] 0.2034235
##
## $`Average Difference in Weeks Between Simulated Study Peak and City Peak`
## [1] -18
```

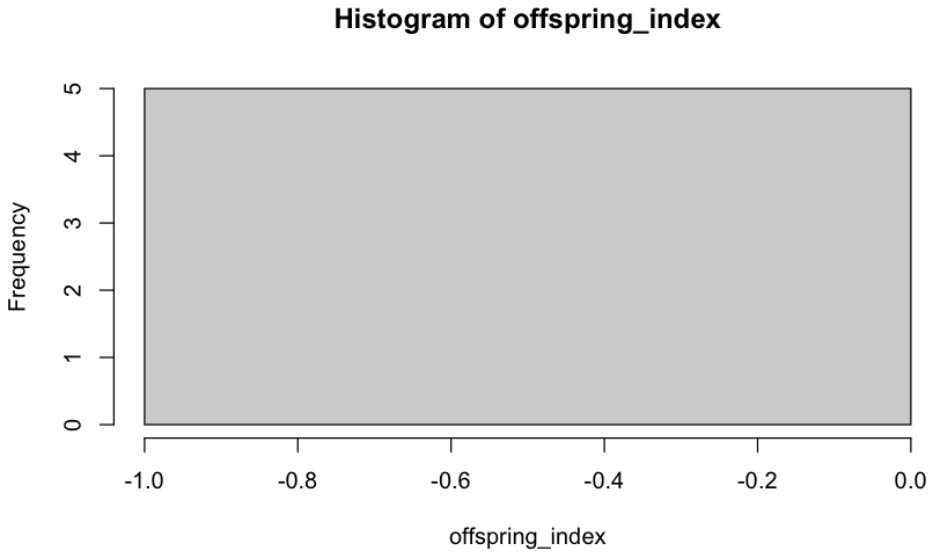


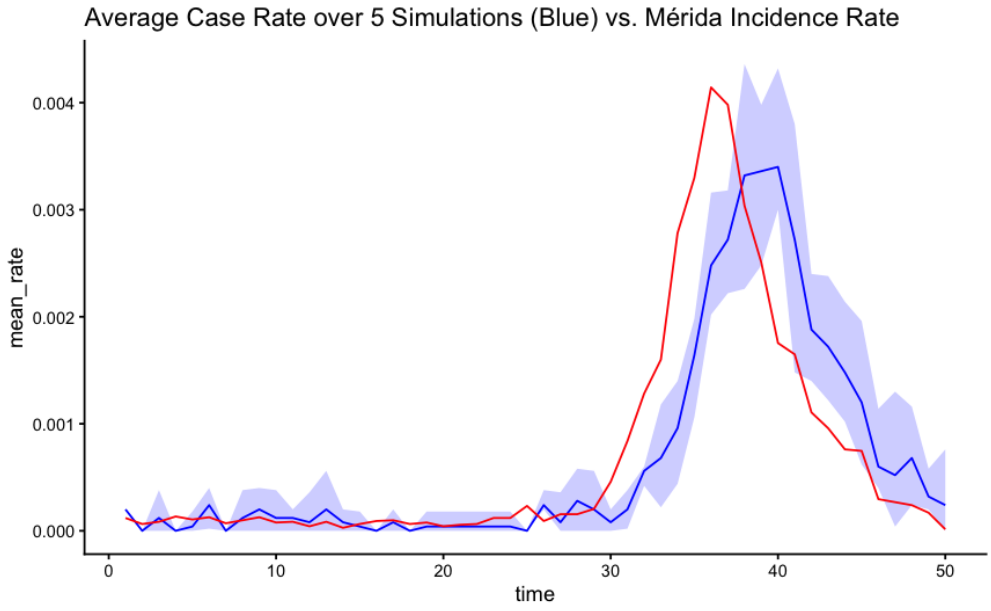


gamma

```
Sys.time()
alpha <- 0; beta <- 0; gamma <- 1; r <- 1
run_sims(5, trial_pop, gen_prob, 1, alpha, beta, gamma, r)
Sys.time()
```

```
## $`Average R_0`
## [1] 0
##
## $`Variance of R_0`
## [1] 0
##
## $`Num Simulations`
## [1] 5
##
## $`Average RR at Peak Incidence`
## [1] 0.9655714
##
## $`Variance of RR at Peak Incidence`
## [1] 0.003496231
##
## $`Average Difference in Weeks Between Simulated Study Peak and City Peak`
## [1] 3.4
```

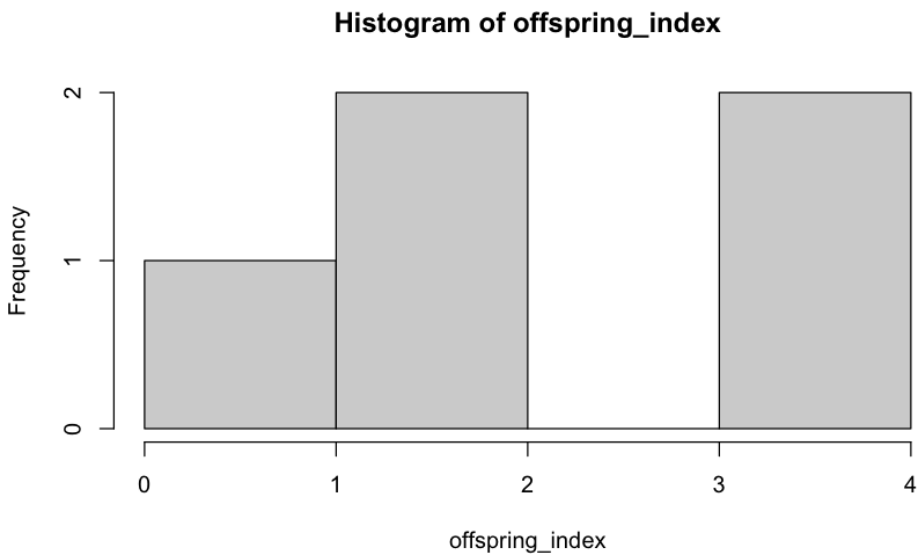


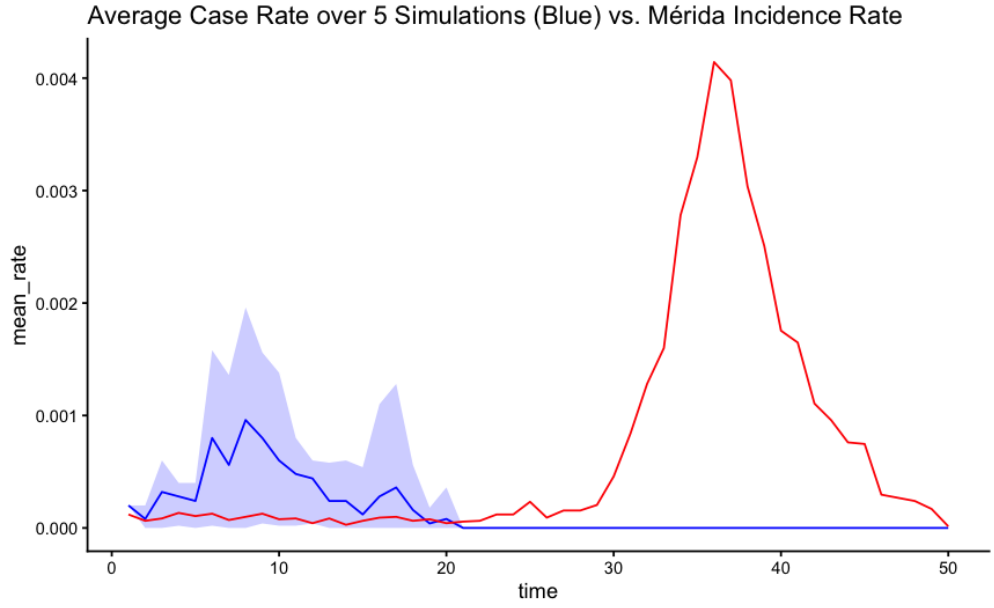


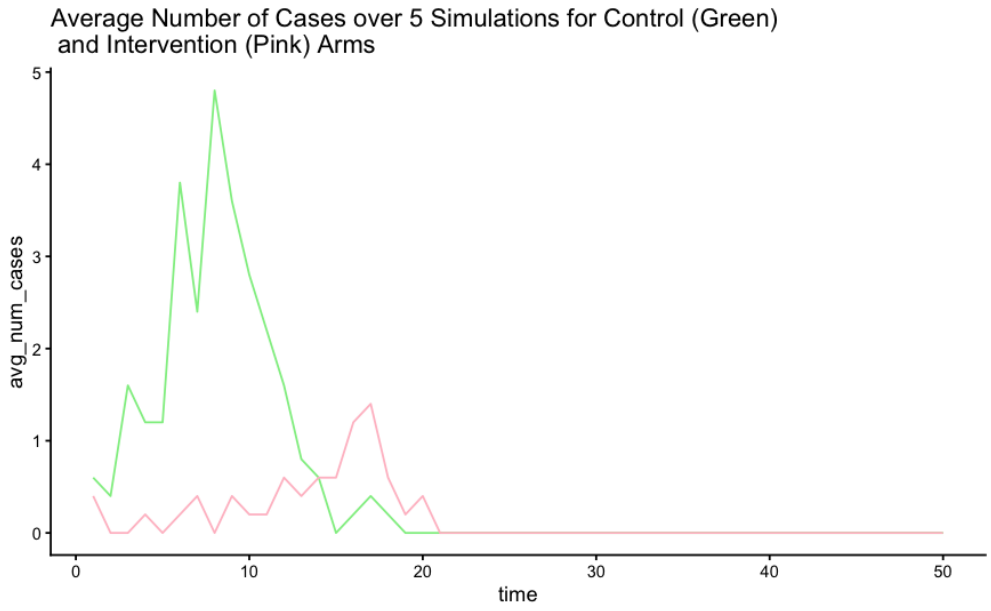
r

```
Sys.time()
alpha <- -log(46/49); beta <- 0; gamma <- 0; r <- 0.5
run_sims(5, trial_pop, gen_prob, 1, alpha, beta, gamma, r)
Sys.time()
```

```
## $`Average R_0`
## [1] 2.4
##
## $`Variance of R_0`
## [1] 2.8
##
## $`Num Simulations`
## [1] 5
##
## $`Average RR at Peak Incidence`
## [1] 0.3282943
##
## $`Variance of RR at Peak Incidence`
## [1] 0.0272706
##
## $`Average Difference in Weeks Between Simulated Study Peak and City Peak`
## [1] -27.8
```


Histogram of offspring_index
Frequency
0
1
2
0
1
2
3
4
offspring_index


Average Case Rate over 5 Simulations (Blue) vs. Mérida Incidence Rate
mean_rate
0.000
0.001
0.002
0.003
0.004
0
10
20
30
40
50
time



Grid Search

parameter grid search (just testing to see if this will run for now)

```
param_grid <- expand.grid(
  alpha = c(0, -log(46/49)),
  beta  = c(0, -log(897/900)),
  gamma = c(0, 1e-6)
)

grid_results <- vector("list", nrow(param_grid))

for (p in 1:nrow(param_grid)) {
  a <- param_grid$alpha[p]
  b <- param_grid$beta[p]
  g <- param_grid$gamma[p]

  sim_out <- run_sims(1, trial_pop, gen_prob, 1, a, b, g, r = 0.5,
                      make_plots = FALSE)

  grid_results[[p]] <- data.frame(alpha = a,
                                  beta = b,
                                  gamma = g,
                                  avg_R0 = sim_out$'Average R_0',
                                  avg_RR_peak = sim_out$'Average RR at Peak Incidence')
}

grid_results_df <- bind_rows(grid_results)
```

```
##       alpha        beta gamma avg_R0 avg_RR_peak
## 1 0.0000000 0.000000000 0e+00      0  0.04827857
## 2 0.0631789 0.000000000 0e+00      1  0.04827857
```

```
## 3 0.0000000 0.003338901 0e+00      0  0.04827857
## 4 0.0631789 0.003338901 0e+00      1  2.22081429
## 5 0.0000000 0.000000000 1e-06      0  0.04827857
## 6 0.0631789 0.000000000 1e-06      2  0.28967143
## 7 0.0000000 0.003338901 1e-06      0  0.04827857
## 8 0.0631789 0.003338901 1e-06      0  0.04827857
```

output function